00:00–01:30 · Nikhil

The next two hours

T2D Glucose Digital Twin

A synthetic subject. A two-hour forecast. An explicit uncertainty.

Akshaya R
BGS College of Engineering and Technology

Nikhil VS
East West Institute of Technology

A reading needs context

What may happen next?

Which context shaped the estimate?

When should the result be withheld?

Our subject is synthetic

293 active synthetic subjects

205 training / 44 validation / 44 test

Demonstration: training subject only

Replay is not a live wearable feed

The forecasting path

01 Existing synthetic CGM and EHR

02 UKF physiological prior

03 PR-TFT forecast and risk

04 FastAPI versioned contract

05 Next.js validation and workspace

A prior and a learned correction

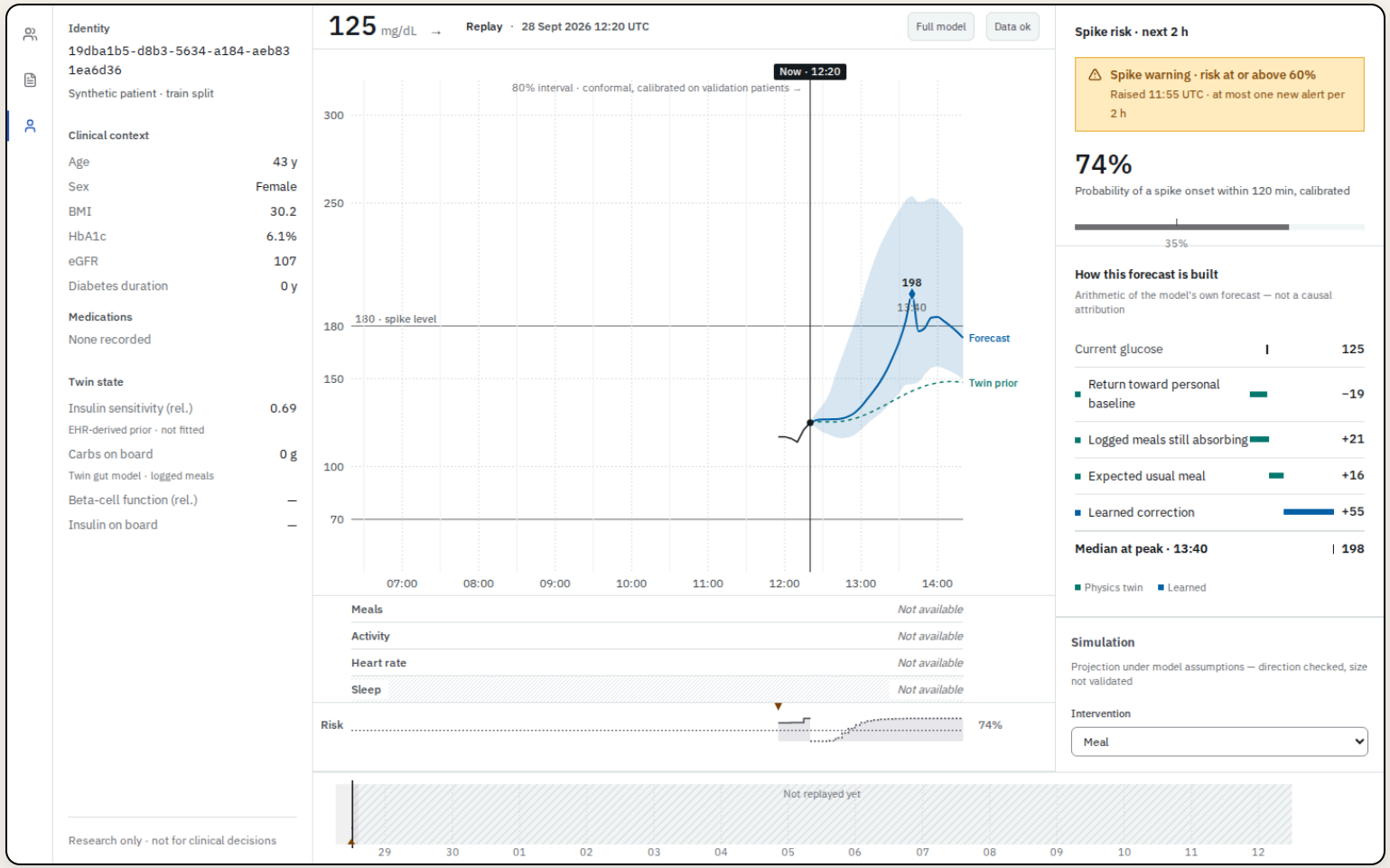
UKF prior: a simplified physiological trajectory

PR-TFT: learned forecasting from observable context

Checkpoint identity does not prove clinical validity

Training manifest records a dirty working tree

Reading a forecast



120-minute horizon, five-minute steps

Median and uncertainty answer different questions

Recorded synthetic replay

Repository-recorded screenshot; provenance in notes. Synthetic research only. No clinical validation claimed.

09:00–11:30 · Nikhil + Akshaya

One subject, successive updates

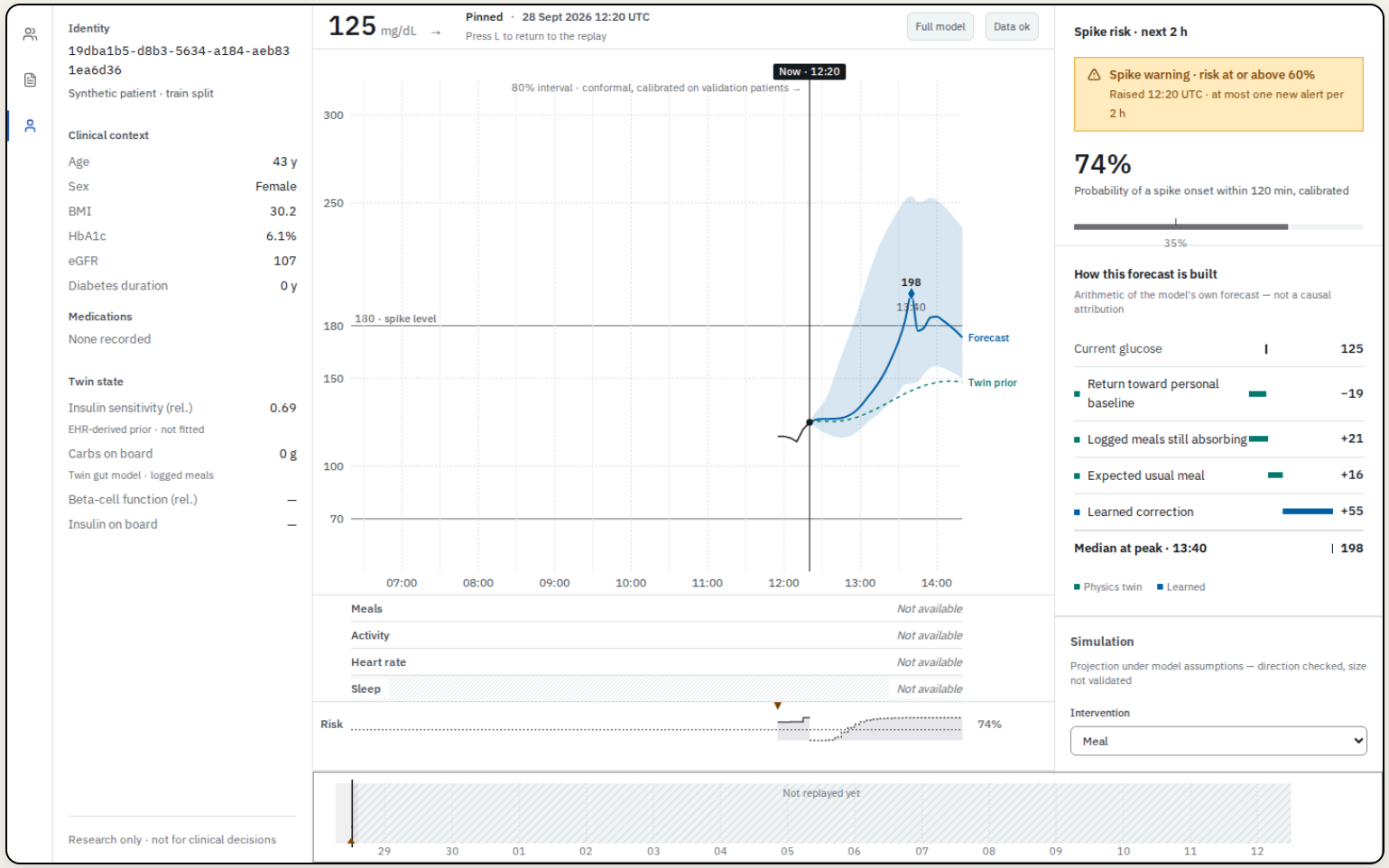
Connected synthetic application

Identity / timestamp / quality / forecast

Forecast updates at recorded timestamps

Synthetic training subject P0003

Holding a moment in time



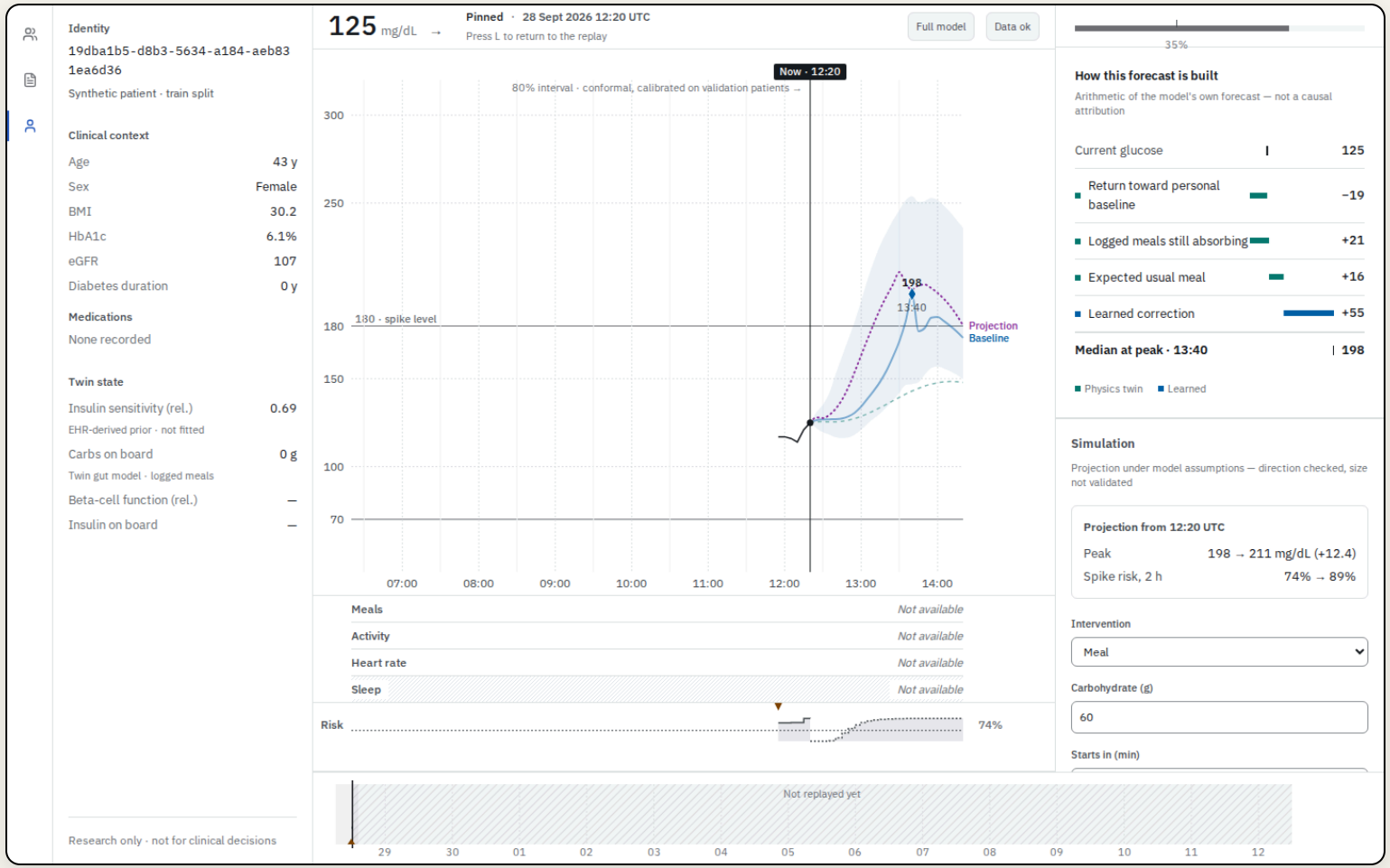
A pinned view keeps the forecast anchor explicit

Timeline navigation uses received history

Recorded repository state, not a new execution

Repository-recorded screenshot; provenance in notes. Synthetic research only. No clinical validation claimed.

Scenario projections



Meal and activity projections exist in source

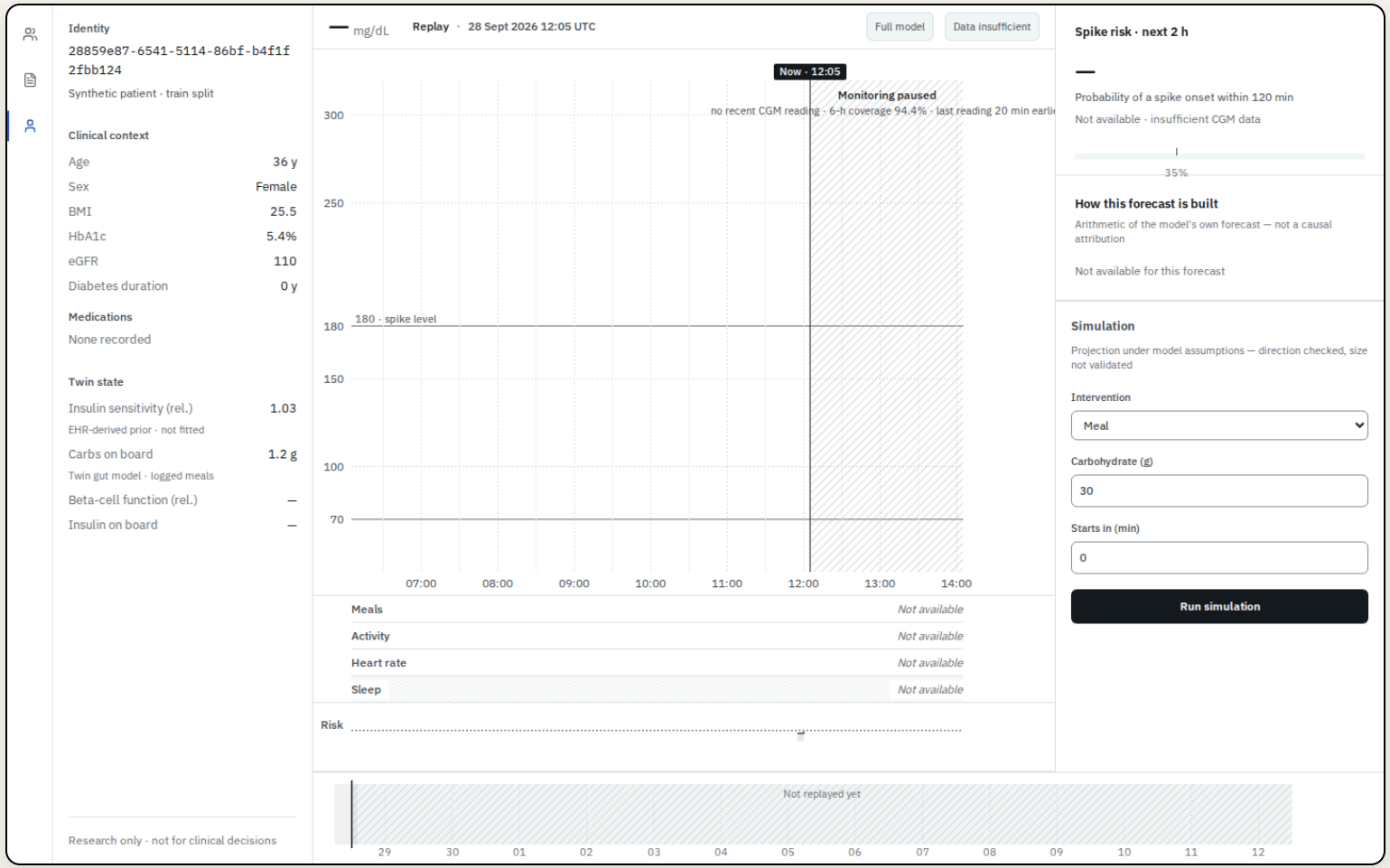
Medication is not modelled

Effect size is not clinically validated

Recorded 60 g example; not treatment advice

Repository-recorded screenshot; provenance in notes. Synthetic research only. No clinical validation claimed.

When the answer is unavailable



Insufficient history can mean no forecast

Stale, disconnected and invalid are distinct states

Reconnect and fallback need further verification

Repository-recorded screenshot; provenance in notes. Synthetic research only. No clinical validation claimed.

What the recorded metrics say

Recorded synthetic validation, not reproduced here

30 / 60 / 120 min RMSE: 12.36 / 19.53 / 25.03 mg/dL

80% interval coverage: 80.00%

Interval width: 37.30 mg/dL

Hazard AUROC: 0.923 | **Brier:** 0.0703

17:30–19:00 · Akshaya

The realism check failed

FAIL

Gate 3 synthetic realism

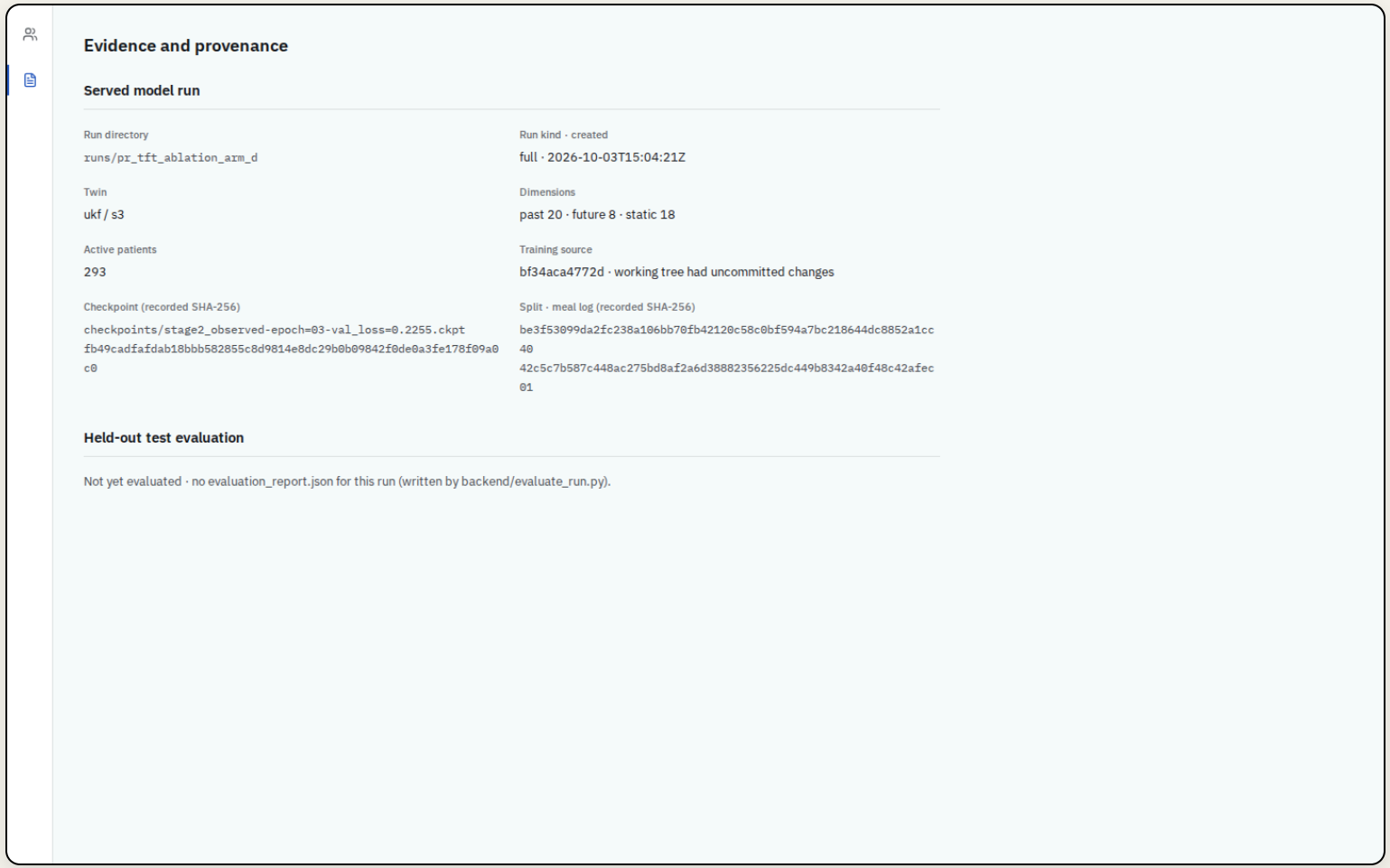
Mock-real comparator

AUROC **0.932** (acceptance target ≤ 0.65)

External EW-01 remains unexecuted

Clinical efficacy is unestablished. Protocol amendments remain pending co-author sign-offs; no human efficacy or cross-center transfer is claimed.

What a reviewer can inspect



Source, contracts and manifest are inspectable

Validation metrics and held-out evaluation differ

Provenance supports review, not clinical efficacy

Repository-recorded screenshot; provenance in notes. Synthetic research only. No clinical validation claimed.

20:30–21:30 · Akshaya + Nikhil

The next two hours, with evidence

An inspectable synthetic forecasting workflow

Next: engineering verification and external validation

Akshaya R and Nikhil VS

Thank you